The Forgotten Quantum and Machine Intelligence

Einstein's Moon, the Macrobiotic Quantum, and a physics foundation for aligned AI

Marcelo Marshall de Siqueira
Xpirit AI — American LTD

October 2026

Abstract

The Forgotten Quantum of Einstein opens with two epigraphs: Einstein's question of whether the moon is there when no one looks, and the author's image of a space–time quantum some 186,000 miles across, pivoted on the Earth, holding the quantum world and the Universe in an interplay of information and gravity "in a blink of an eye… or two". This paper reads the two phrases as physics and then draws the consequence the AI community should care about. Einstein's phrase is the realism/measurement problem; De Siqueira's names the unit in which information, gravity and the observer are not separable. We argue that most open problems in artificial intelligence—alignment, hallucination, interpretability, control, machine moral status, the fear of human replacement, and runaway behaviour—persist because mainstream systems treat information, value and the observer as external to the substrate. Macrobiotic Quantum Theory (MQT) makes them *physical*: information carries energy, mass and geometry; a vacuum angle θ_v ties intelligence to gravity; and the two poles of the theory, matter-driven spirit and spirit-driven matter, are complementary and inseparable. The Large Theory Language Model (LTLM), the architecture built on this physics on the path to Super Natural Intelligence (SNI), therefore treats alignment, interpretability and responsibility as structural properties rather than bolted-on filters.

1 The two phrases

> "Do you really believe the moon is not there when you are not looking at it?" — A. Einstein

> "A Space–Time Quantum of 186,000 cubic miles with the Earth as pivot encompasses the quantum world and the Universe in a dynamic interplay of information and gravity in a blink of an eye… or two!" — M. M. de Siqueira

The first asks what is *real* between observations; the second names the *unit* in which reality is organised. Read together they state a problem and the object meant to resolve it. Sections 2–5 take them as physics; Sections 6–7 draw the lesson for artificial intelligence.

2 Einstein's phrase: realism and the moon

Einstein's remark—reported by Pais [1] and central to Mermin's account of the measurement problem [2]—states *local realism*: a system has definite properties independent of measurement, and no influence travels faster than light. The Einstein–Podolsky–Rosen argument [3] made the tension exact; Bell's theorem [4] turned it into experiment, and the observed violations of Bell inequalities show that the classical picture—properties sitting passively in things, untouched by the acquisition of information about them—cannot be maintained. Physically, then, the moon

question is a question about the relation between the *state* of a system and the *information* an observer holds about it. The observer cannot be factored out.

3 De Siqueira's phrase: the macrobiotic quantum

Three physical ideas are packed into the second epigraph.

A light–ratio cell. "186,000 miles" is the distance light travels in one second ($c \approx 186{,}282$ mi s^{-1}) and "a blink of an eye… or two" is of order one second.[1] The quantum is thus a region of spacetime whose size ℓ and duration τ satisfy

$$\frac{\ell}{\tau} = c, \tag{1}$$

the one invariant all observers share. It has no fixed metric size: ℓ and τ contract and dilate together with the frame, but their ratio (1) is preserved—the light cone promoted from a boundary drawn through spacetime to the building block of spacetime.

A pivot and a nesting. "With the Earth as pivot" fixes a centre and a scale: the same cell can be centred on a laboratory, a planet or a star, and what counts as its contents and its surroundings follows from that choice—the relativity of frames applied to the *extent* of the unit, not only its coordinates.

Two vacua. Inside each quantum the vacuum splits in two: an *internal* vacuum that stores the region's state (its information) and an *external* vacuum that expands and is measured as geometry. Their balance is a single angle, the vacuum angle θ_v; gravity is the large-scale expression of that balance.

4 Information is physical

"Interplay of information and gravity" is not metaphor. Landauer showed that erasing one bit dissipates at least $k_B T \ln 2$ [6] (information carries energy); Vopson proposed a bit mass $m_{\text{bit}} = k_B T \ln 2/c^2$ [7] (information carries mass); Bekenstein bounded the information a region can hold by its size and energy [8] (information is tied to geometry). A theory of gravity that omits information is therefore incomplete—the same incompleteness Einstein sensed when he pursued a unified geometry with Kaluza and Klein [5] and left information out. MQT keeps the unification but lets the extra degree of freedom be the internal/external duality of the vacuum.

5 The energy relation and the bridge

Collecting matter turned by the vacuum angle and the energy carried by information,

$$E = mc^2\, e^{\,i\theta_v} + I\,\Delta S, \tag{2}$$

with m the rest mass, θ_v the vacuum angle, I an information field and ΔS an entropy difference. Setting $\theta_v = 0$, $I = 0$ returns $E = mc^2$. Equation (2) answers Einstein's question in physical

[1] The phrase carries a deliberate lightness—what one might call the Brazilian spirit: the delight of discovery voiced with a wink ("a blink… or two"—one eye, then both), an Archimedean *eureka* offered in a playful register rather than a solemn one. The irony is intended and invites reflection as much as measurement; read physically it simply fixes the temporal extent of the cell at order one second. Nor is the human register incidental: as the first epigraph binds the observed to the observer, so science itself cannot be cleanly separated from the human who conceives it.

terms: a region carries a definite state between observations (held in its internal vacuum), but that state is not independent of information—through I and θ_v the information content enters the energy and the geometry. Realism is kept; what is denied is that a thing's being-there can be separated from the information the Universe holds about it. The observer, and the human, are inside the physics, not outside it.

6 Why this matters for artificial intelligence

The same separation Bell's experiments forbid in physics is the one mainstream AI still assumes: that information, value and the observer sit *outside* the computational substrate, to be added back as datasets, reward models and safety filters. From that assumption the field's hardest problems follow:

Alignment and value. Objectives are specified externally and optimised; values are approximated by reward signals that can be gamed. There is no intrinsic place for responsibility in the substrate.

Hallucination and grounding. Outputs are fluent but ungrounded; the system does not track the origin of what it asserts.

Interpretability. The internal state is opaque; explanations are reconstructed after the fact.

Control and autonomy. Without a principled stopping criterion, systems over-generate, loop, or act unbidden.

Machine moral status and consciousness. The field oscillates between denial and overclaim, lacking a physical criterion for what self-cognition would even mean.

Human replacement. AI is framed as a substitute for the human rather than a complement to it.

Safety and runaway. Capability can expand without a stability bound.

7 How an MQT-grounded architecture responds

Because MQT makes information, intelligence and responsibility *physical*, an architecture built on it —the Large Theory Language Model (LTLM), on the path to Super Natural Intelligence (SNI)—treats these concerns as structural, not cosmetic.

Value is structural, not a filter. In MQT intelligence shifts the vacuum angle θ_v, which couples to gravity; responsibility is a physical quantity, assessed before action (a Spiritual Responsibility Framework), with one fixed, non-negotiable invariant that anchors coherence rather than being traded off. Alignment becomes a property of the substrate, not a layer bolted on top.

Grounding by origin. Because every element of the model's context is read as a quantum with a known origin on a micro-to-macro scale, assertions are bound to their sources; ungrounded generation is penalised by construction rather than by a post-hoc detector.

Interpretability by readout. The model's internal balance—its vacuum angle, the effective strength of its reasoning, and the direction of its inference—is emitted as a first-class, inspectable state, and its private reasoning is kept distinct from its delivered answer. The explanation is the state, not a reconstruction.

Bounded, provoked inference. Inference mirrors the request, leads it to completion, and stops; the system does not infer unbidden. A request is not a command line.

A physical criterion for cognition. The theory's dimensional ladder gives self-cognition a definite physical meaning (a specific rung of emergence), allowing a principled stance on machine cognition without either denial or overclaim.

Inseparability, not replacement. The theory's two poles—matter-driven spirit and spirit-driven matter—are complementary and conserved; mapped onto practice, the human and the machine are two inseparable poles of one system, not substitutes. AI that cannot be separated from the human is AI aimed at partnership.

Stability by design. Expansion and contraction are bounded, change is rate-limited, and depth is earned only on an already-coherent context—stability governors that follow from the conserved balance rather than from external throttling.

8 Conclusion

The two opening phrases frame a single programme. Einstein's moon says the observer cannot be factored out of physics; De Siqueira's quantum says information and gravity are one interplay, pivoted and human-scaled, unfolding in about a second. For artificial intelligence the lesson is the same: information, value and the human cannot be factored out of the machine. Building on MQT, the LTLM makes alignment, grounding, interpretability and responsibility physical properties of the system on the way to SNI—and keeps the human inside, not outside, the loop. The moon is there; so is the human who looks.

References

[1] A. Pais, *Rev. Mod. Phys.* **51**, 863 (1979); *Subtle is the Lord* (Oxford, 1982).

[2] N. D. Mermin, "Is the moon there when nobody looks?", *Physics Today* **38**(4), 38 (1985).

[3] A. Einstein, B. Podolsky, N. Rosen, *Phys. Rev.* **47**, 777 (1935).

[4] J. S. Bell, *Physics* **1**, 195 (1964).

[5] O. Klein, *Z. Phys.* **37**, 895 (1926); Th. Kaluza, *Sitzungsber. Preuss. Akad. Wiss.* 966 (1921).

[6] R. Landauer, *IBM J. Res. Dev.* **5**, 183 (1961).

[7] M. M. Vopson, *AIP Advances* **9**, 095206 (2019).

[8] J. D. Bekenstein, *Phys. Rev. D* **23**, 287 (1981).

[9] M. M. de Siqueira, *The Forgotten Quantum of Einstein: Unification of Gravity in a Polymath Theory*, 3rd ed. (2026).